UI/UX Design and Development Planning Report

AI Code Security Analyzer

CSE4104 – Software Engineering Lab

Section: 7C

Team: 04

Submitted To

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1. Team Information

Field	Information
Team Name	CSE4104-7C-T04
Section	7C
Project Title	AI Code Security Analyzer
Team Leader	Sadia Islam Jhumur
GitHub Repository	`https://github.com/0xSaikat/pluto-ai`

Table 1: Project Information

Student ID	Member Name
11230121171	Sadia Islam Jhumur (Team Leader)
11230121188	Sakil Hasan Saikat
11220121034	Tanjilanta Kanchi
11230121061	Farin Islam Mim

Table 2: Team Members

2. Project Overview

AI Code Security Analyzer is an AI-powered web application developed to assist developers and organizations in identifying security vulnerabilities within source code. The platform leverages modern Artificial Intelligence technologies to analyze source code, assess security risks, and generate professional vulnerability reports.

The system aims to simplify secure software development by providing automated vulnerability detection, risk assessment, security reporting, and compliance verification through an intuitive web interface.

Core Features

- AI Vulnerability Detection
- Secure Code Analysis
- Risk Assessment
- Security Report Generation
- Real-Time Monitoring
- Compliance Verification

3. UI Design Screens

Screen 1: Home Page (Initial Version)

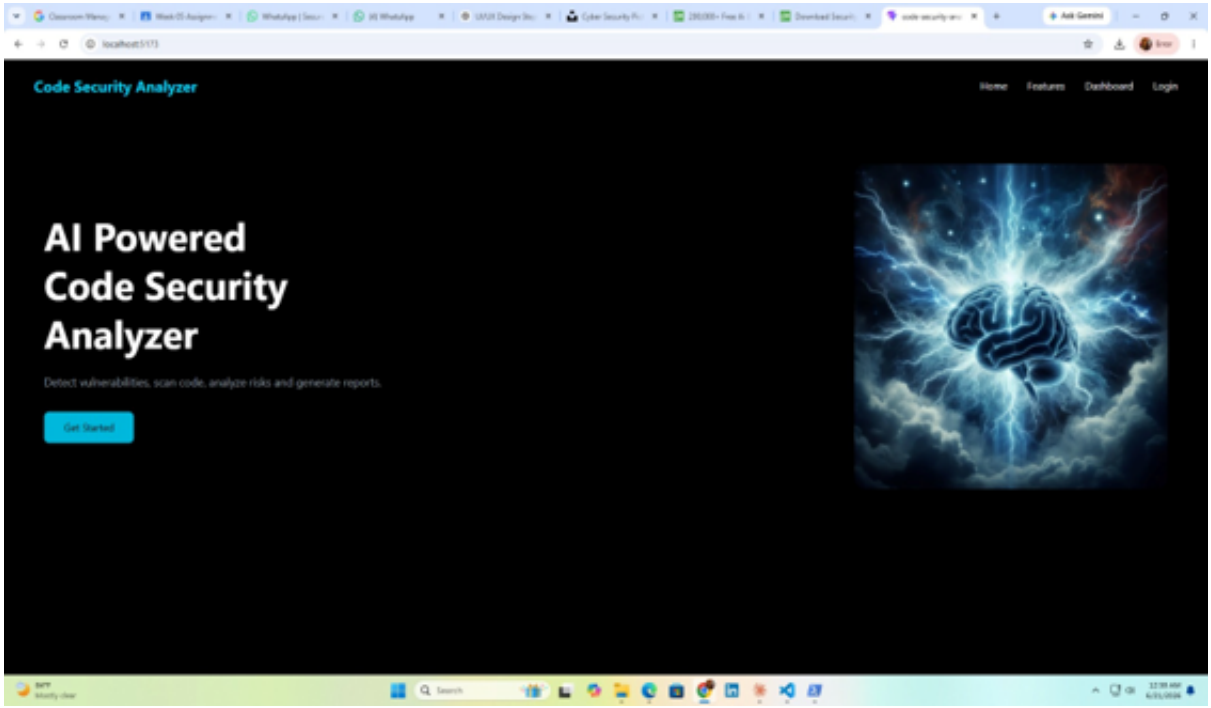


Figure 1: Home Page (Initial Version) of the AI Code Security Analyzer.

Design Analysis

The Home Page serves as the entry point of the application and introduces users to the platform. The interface utilizes a cybersecurity-inspired dark theme combined with blue accent colors to establish a professional and security-focused visual identity. The prominent "Get Started" button guides users toward registration and platform usage.

Screen 2: Home Page (Refined Version)

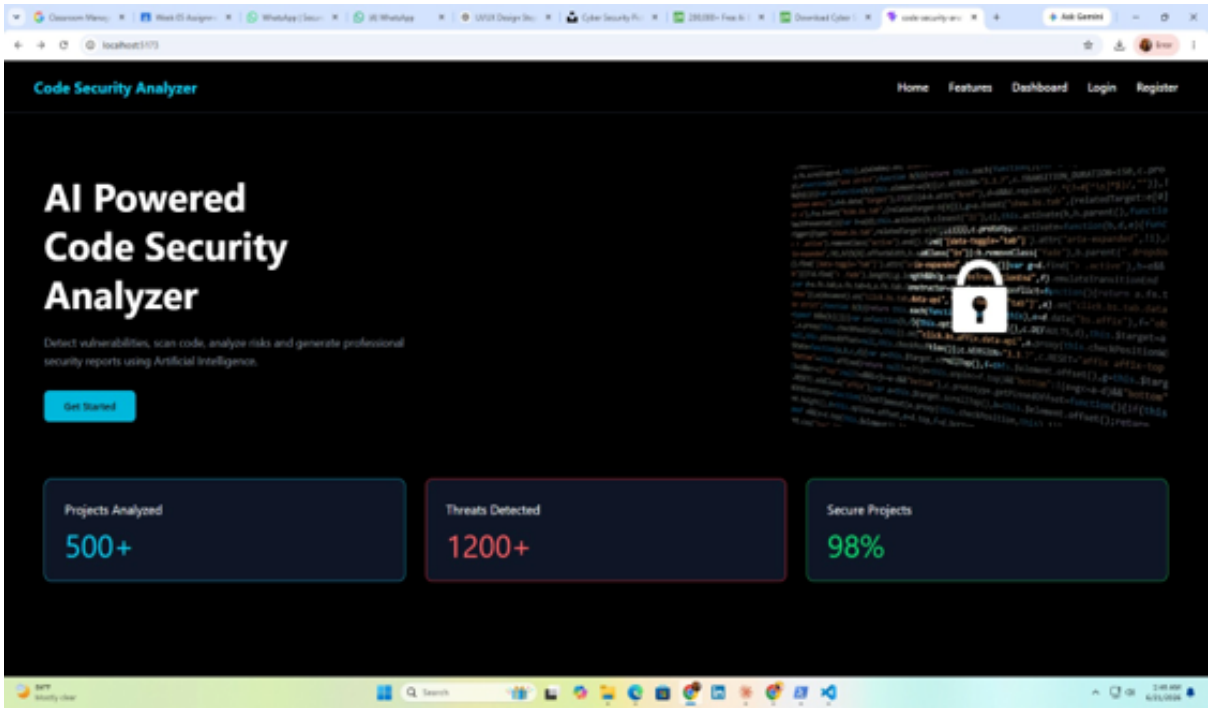


Figure 2: Refined Home Page displaying platform statistics and enhanced visual design.

Design Analysis

This improved version of the landing page introduces platform statistics including analyzed projects, detected threats, and secure projects. These metrics provide users with immediate insight into platform effectiveness while maintaining a clean and responsive layout.

Screen 3: Features Page

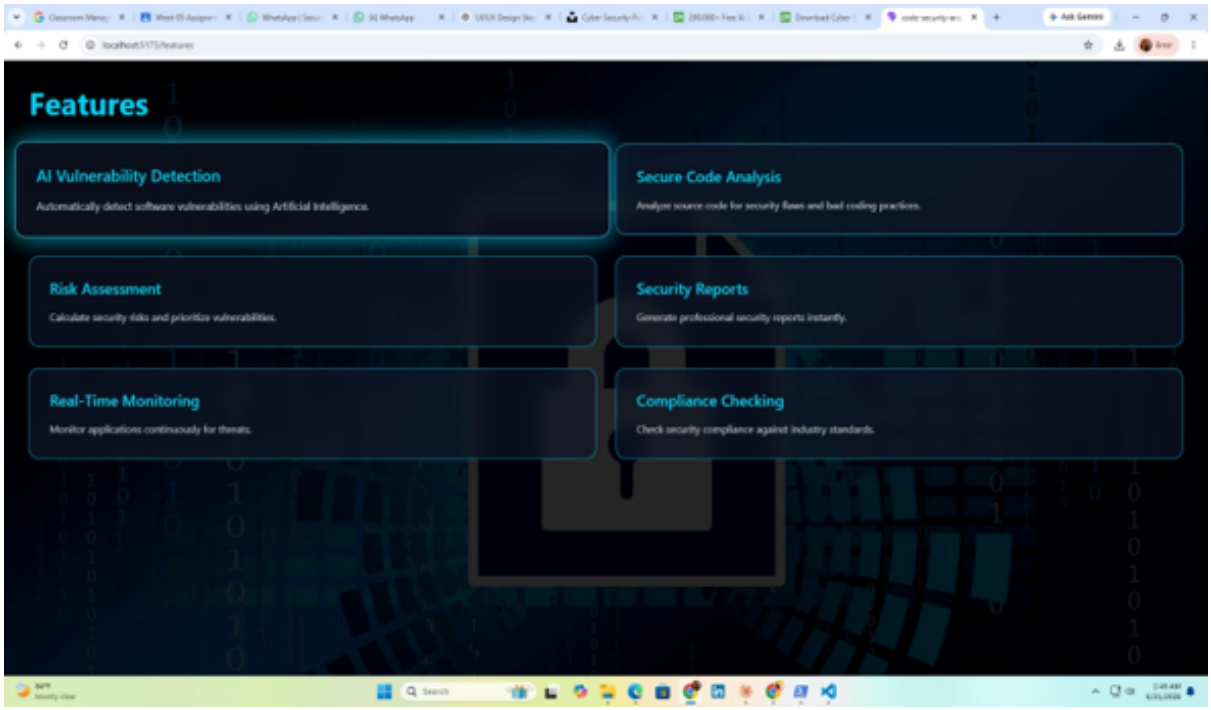


Figure 3: Features Page highlighting the core functionalities of the platform.

Design Analysis

The Features Page presents the primary capabilities of the system through a card-based layout. The page highlights AI vulnerability detection, secure code analysis, risk assessment, report generation, real-time monitoring, and compliance verification. The layout improves readability and enables users to understand the system capabilities quickly.

Screen 4: User Dashboard

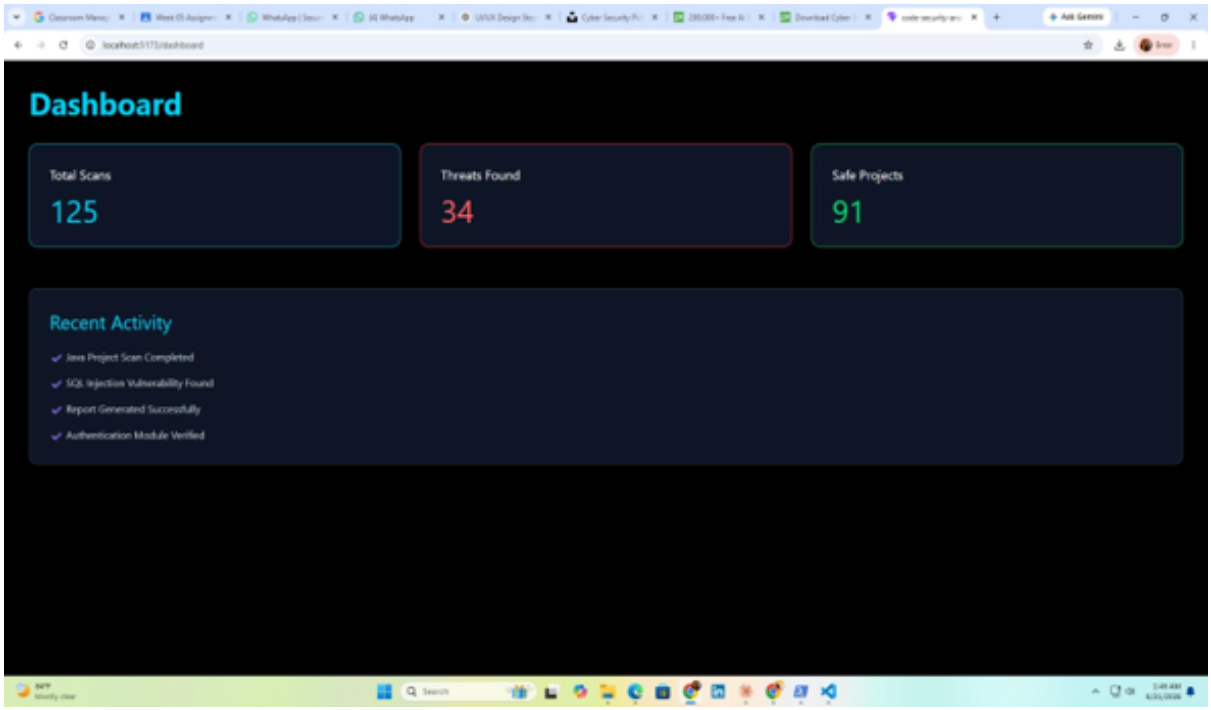


Figure 4: User Dashboard showing scan statistics and recent security activities.

Design Analysis

The Dashboard serves as the primary workspace for authenticated users. It provides an overview of security scan statistics, detected threats, safe projects, and recent activity logs. The dashboard allows users to monitor project security status efficiently from a centralized interface.

Screen 5: Login Page

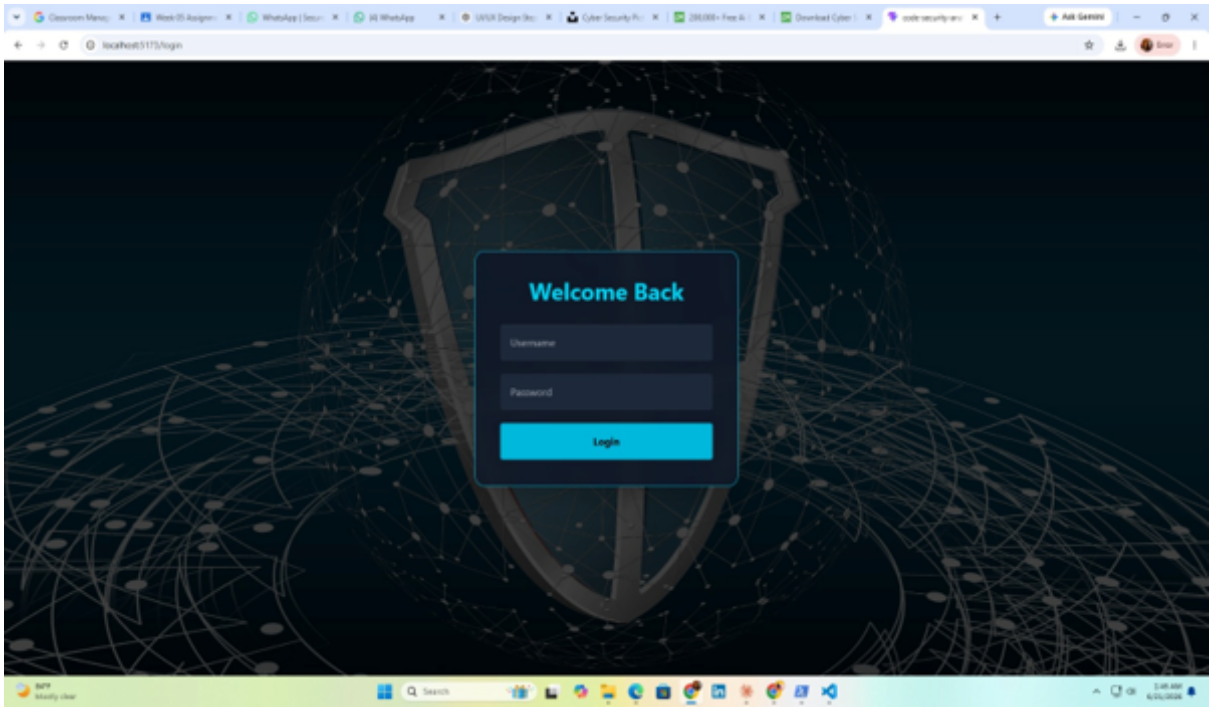


Figure 5: Secure Login Page used for user authentication and access control.

Design Analysis

The Login Page provides a secure authentication mechanism that enables registered users to access protected resources within the system. The shield-themed background reinforces the security-focused nature of the platform while maintaining visual consistency with the overall design.

Screen 6: Registration Page

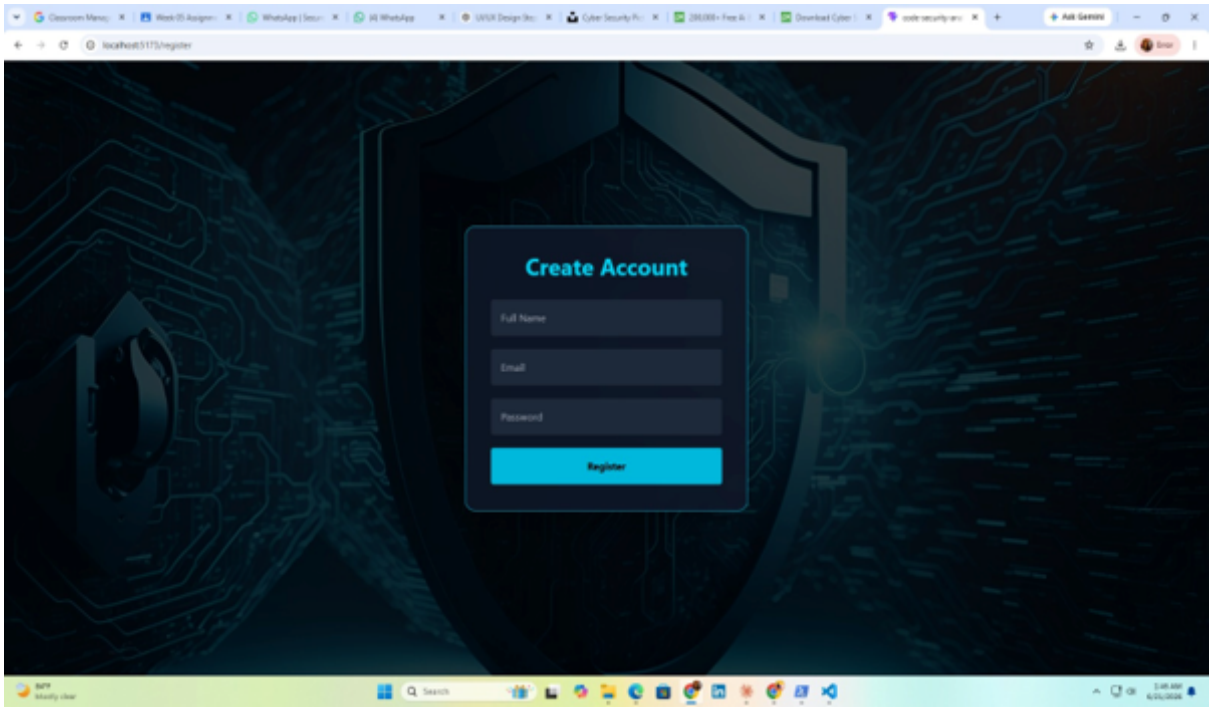


Figure 6: Registration Page allowing new users to create platform accounts.

Design Analysis

The Registration Page enables guest users to create accounts and access the AI-powered security analysis services. The design follows the same visual theme as the Login Page, ensuring a consistent user experience throughout the authentication process.

4. User Flow Design

Landing Page

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Registration

↓

Login

↓

Dashboard

↓

Upload Source Code / Submit Repository

↓

AI Security Analysis

↓

Vulnerability Detection

↓

Results Display

↓

Security Report Generation

↓

Report Download

The user flow illustrates how users interact with the AI Code Security Analyzer platform. Beginning from the landing page, users register and authenticate themselves before accessing the dashboard. They can then submit source code for analysis, review vulnerability findings, and generate downloadable security reports.

5. Development Roadmap

Week	Phase	Activities
06	Backend Development	Database schema design, API development, authentication, and code upload functionality
07	Frontend Development	React components, routing implementation, and responsive user interface
08	AI Integration	Integration of GPT-4o, Gemini, Claude, and DeepSeek APIs
09	Feature Completion	Risk assessment, report generation, and compliance checking modules
10	Testing and Debugging	Unit testing, integration testing, performance optimization, and bug fixing
11	Deployment	Cloud deployment, documentation finalization, and project presentation preparation

Table 3: Development Roadmap

6. Team Task Distribution

Member	Responsibilities
Sadia Islam Jhumur	Project Management, UI/UX Design, and Documentation
Sakil Hasan Saikot	Backend Development, API Implementation, and Authentication System
Tanjilanta Kanchi	Database Design, Report Generation, and Data Management
Farin Islam Mim	Frontend Development, Testing, and Quality Assurance
All Members	Integration Testing, Deployment, Final Review, and Presentation

Table 4: Team Responsibilities

7. Final Technology Stack

Category	Technology
Frontend	React.js, Vite, Tailwind CSS
Backend	FastAPI (Python)
Database	PostgreSQL, MongoDB
AI Services	GPT-4o, Gemini, Claude, DeepSeek
Authentication	JWT Authentication
Deployment	Vercel, Render
Version Control	Git and GitHub

Table 5: Final Technology Stack

GitHub Repository

`https://github.com/0xSaikat/pluto-ai`